AWS Lab: Cloud Service Model Classification & Shared Responsibility Boundary

Duration : 30 Min.

Scenario

An enterprise migration plan mistakenly designates AWS RDS PostgreSQL as IaaS, asserting that the internal infrastructure team is responsible for patching the guest Linux operating system. The cloud engineer must verify the guest OS access model for RDS, clarify the shared responsibility boundary, and generate a service classification document.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** Amazon Web Services (AWS) - **Managed Database Engine:** Amazon RDS for PostgreSQL - **Compute Engine:** Amazon EC2 (Elastic Compute Cloud) - **Access Boundary:** No OS/root SSH access on RDS; Full guest-OS root access on EC2

Username & Naming Conventions

Your candidate prefix is: `<prefix>` (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Inspect AWS RDS database configuration and verify that no SSH/guest-OS shell is exposed. 2. Reclassify Amazon EC2 as IaaS and Amazon RDS as PaaS. 3. Configure and verify in the AWS environment customer vs AWS patching responsibilities.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Service classification: accurately identify RDS as PaaS and EC2 as IaaS	4 Marks
TC2	Responsibility mapping: map OS patching, hypervisor security, and data backup across the boundary	3 Marks
TC3	JSON output: aws_shared_responsibility.json conforms to expected schema	3 Marks

Total Score: 10 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.